

Civil Engineering

Vectors, Matrices and Engineering Data Analysis

Revision Booklet

Use these questions for CAT, assignment preparation and summative revision.

Formula Recall

1. State where this formula is applied and create one original example.

$$\vec{R} = \vec{A} + \vec{B}$$

2. State where this formula is applied and create one original example.

$$A_x = A \cos \theta$$

3. State where this formula is applied and create one original example.

$$A_y = A \sin \theta$$

4. State where this formula is applied and create one original example.

$$AX = B$$

5. State where this formula is applied and create one original example.

$$s = \sqrt{\frac{\sum (x - \bar{x})^2}{n-1}}$$

Structured Revision

1. A realistic civil engineering task involves survey displacement. Formulate the calculation, solve it, and interpret your result.
2. A realistic civil engineering task involves matrix equations. Formulate the calculation, solve it, and interpret your result.
3. A realistic civil engineering task involves quality control. Formulate the calculation, solve it, and interpret your result.
4. A realistic civil engineering task involves statistical judgement. Formulate the calculation, solve it, and interpret your result.
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References and Learning Sources

These learner materials are original notes prepared for Mr Mwirigi Mathematics Training Hub. The topic sequencing and level are informed by standard engineering mathematics references and the uploaded curriculum/OS materials. No textbook pages or solution-manual pages have been reproduced.

1. John Bird, Engineering Mathematics, Newnes / Elsevier.
2. K. A. Stroud and Dexter J. Booth, Engineering Mathematics, Palgrave Macmillan.
3. Uploaded KNEC solved mathematics materials for examination-style orientation.
4. Civil Engineering, Building Technology and Land Survey curriculum and occupational-standard materials used for local alignment.